

Speech-Recognition

Exercises about Intro - Linguistics

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Abstract

Practical introduction to Chatbot project, Theory about terms of Linguistics

Introduction

In the exercises and mid-terms we will extend a chatbot with some functions. As a starting point for the chatbot you may use your development from a previous semester or the one from code-02-chatbot.py . The chatbot must be a single page Streamlit app with voice *and* text input.

Practical Part 01

Setup and test the Python chatbot environment. See the following code in the `material` folder:

- `code_01_check_env.py`
- `code_02_chatbot.py`

Important:

- 1) The chatbot app is a streamlit app that uses microphone and loudspeakers. Make sure that this hardware works correctly on your computer.
- 2) For the development use a Python virtual environment.
- 3) Also make sure, that the streamlit app can work with and store static content. For this, a `.streamlit` folder must exist in the app folder with a file called `config.toml`. The config file must have an entry like this:

```
[server]
enableStaticServing = true
```

Preparation

Step 1: Setting up the Development Environment

Make sure you have Python and (at least) the following libraries installed:

- `base64`
- `librosa` (for MFCCs and diagrams)
- `matplotlib`
- `numpy`
- `pyttsx3`
- `scikit - learn`
- `scipy`
- `soundfile` (for loading WAV files)
- `speech_recognition`
- `streamlit`

Installation via pip:

```
pip install base64 librosa matplotlib numpy pytsx3 scikit-learn scipy soundfile  
SpeechRecognition streamlit
```

Step 2: Download the Speech Commands Dataset

The dataset is large (download size, approx. 2 GB, decompressed files 8GB). You can download it directly from the TensorFlow Datasets page:

http://download.tensorflow.org/data/speech_commands_v0.02.tar.gz

Extract it to a directory of your choice, e.g., './speech_commands_v0.02'

Practical Part 02

Make yourself familiar with the speech commands and the recording of sounds either with audio tools like Audacity or similar.

Analyze the files. Hear the commands. What are the differences between them?

Get information about the file (type, bitrate, etc.).

Theory

Part 01

Describe the difference between Syntax, Semantics and Pragmatics in linguistic studies. Explain an example for each.

Part 02

Break down the following words into their morphemes and identify whether they are free or bound morphemes, and what type of morpheme they are (lexical/grammatical, prefix/suffix/root):

- a) unhappy
- b) dogs